IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

EIE PART II: MEng, BEng and ACGI

FEEDBACK SYSTEMS

Thursday, 28 May 10:00 am

Time allowed: 1:30 hours

There are THREE questions on this paper.

Answer ALL questions. Question 1 carries 40% of the marks. Questions 2 and 3 carry equal marks (30% each).

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : F. Teng

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1. Consider the feedback system in Figure 1.1. Here K represents a constant controller and $G(s) = G_1(s)G_2(s)$ is the system, where each of $G_1(s)$ and $G_2(s)$ is a transfer function that represents the circuit in Figure 1.2. The values of the parameters for $G_1(s)$ are such that $C_i = 0$, $R_iC_f = 1$, $R_f = \infty$, and those for $G_2(s)$ are such that $C_i = 0$, $R_iC_f = 1$, $R_i = R_f$. All the capacitors are assumed to be initially uncharged.

a) Evaluate $G(s)$. [5]

b) Derive the differential equation relating $u(t)$ to $y(t)$. [5]

c) Draw the Nyquist diagram for $G(s)$. [5]

d) Derive a state–space realisation for $G(s)$ in the controllable form. [5]

e) Let $x(t)$ be the state vector in Part (d) above and let $u(t) = -\begin{bmatrix} k_1 & k_2 \end{bmatrix} x(t)$ be a state feedback law. Find k_1 and k_2 such that the closed loop poles are located at -1 and -2. [5]

f) Let $r(t) = 1 - e^{-t}$ applied at $t = 0$. What is the steady–state value of $e(t)$? [5]

g) Let $r(t)$ be a unit step applied at $t = 0$. Find the value of K such that the closed–loop response has a settling time of approximately 8 seconds and a maximum overshoot of approximately 5%. [5]

h) Let $r(t)$ be a unit ramp applied at $t = 0$. Use the final value theorem of the Laplace transform, which you should state, to find the steady–state value of $e(t)$. Give your answer in terms of K. [5]

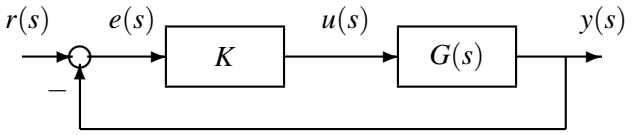


Figure 1.1

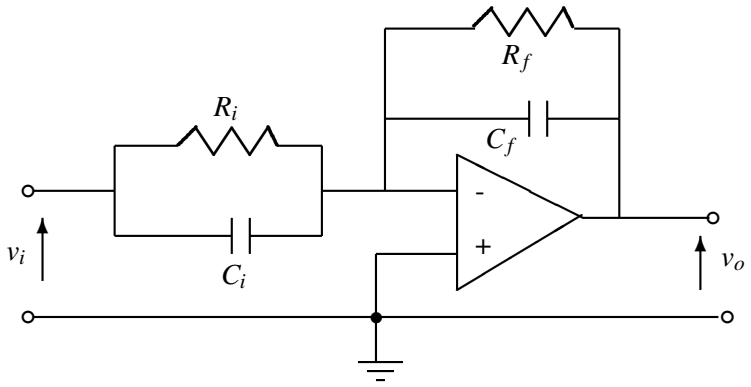


Figure 1.2

2. Consider the feedback system in Figure 2.1.

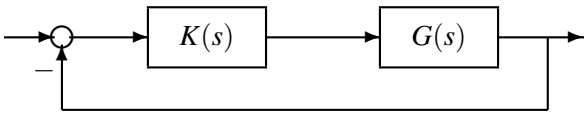


Figure 2.1

Here, $K(s)$ is the transfer function of a controller and $G(s)$ is a stable transfer function which has no finite zeros and whose frequency response is shown in Figure 2.2.

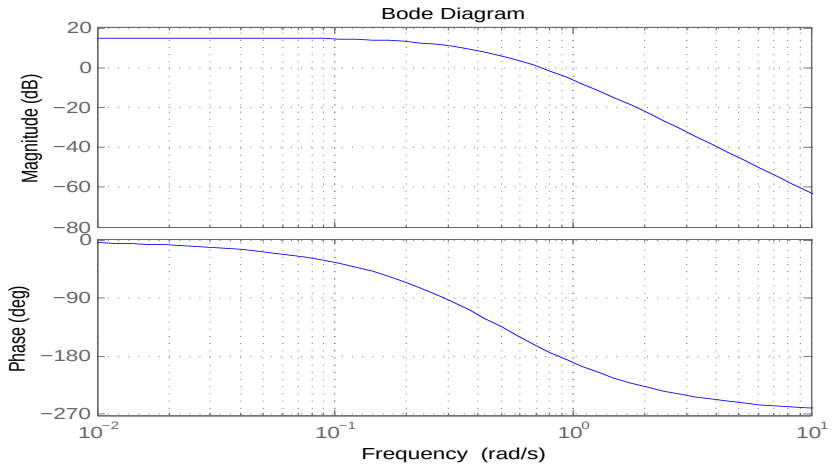


Figure 2.2

a) Use the frequency response above to sketch a rough Nyquist diagram of $G(s)$. Indicate the low and high frequency parts and the real–axis intercepts. [5]

b) Give approximate values for the gain and phase margins and the cross-over frequency. [5]

c) State the Nyquist stability criterion and use it to determine the number of unstable closed-loop poles when (i) $K(s) = 0.5$, (ii) $K(s) = 5$ and (iii) $K(s) = -0.5$. [5]

d) Let $K_1(s)$ have the frequency response shown in Figure 2.3 on page 3. Briefly describe $K_1(s)$ and indicate its effects on the performance and stability of the feedback loop. [5]

e) Let $K_2(s)$ have the frequency response shown in Figure 2.4 on page 4. Briefly describe $K_2(s)$ and indicate its effects on the performance and stability of the feedback loop. [5]

f) Let $K_3(s)$ have the frequency response shown in Figure 2.5 on page 5. Briefly describe $K_3(s)$ and indicate its effects on the performance and stability of the feedback loop. [5]

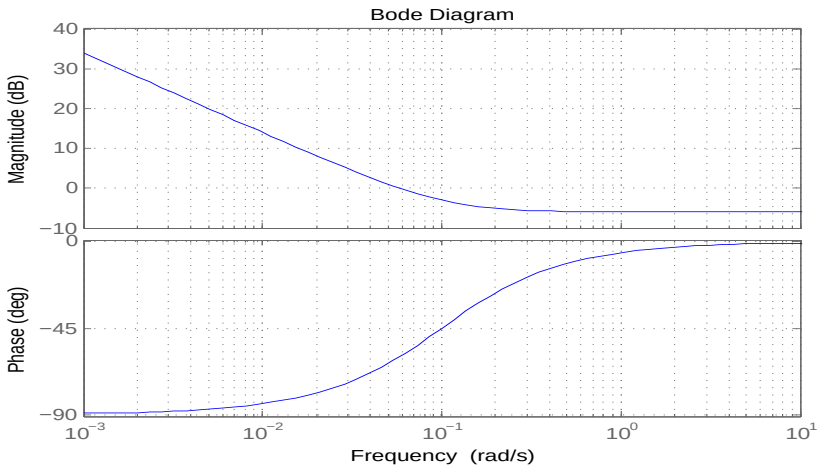


Figure 2.3: Bode plot for $K_1(s)$

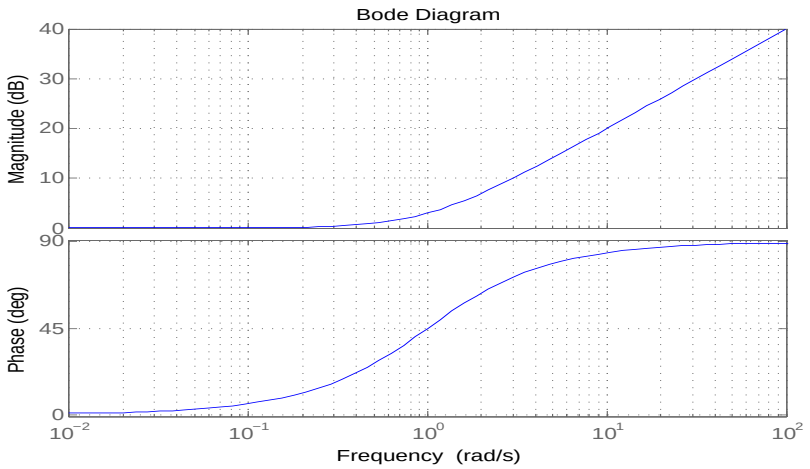


Figure 2.4: Bode plot for $K_2(s)$

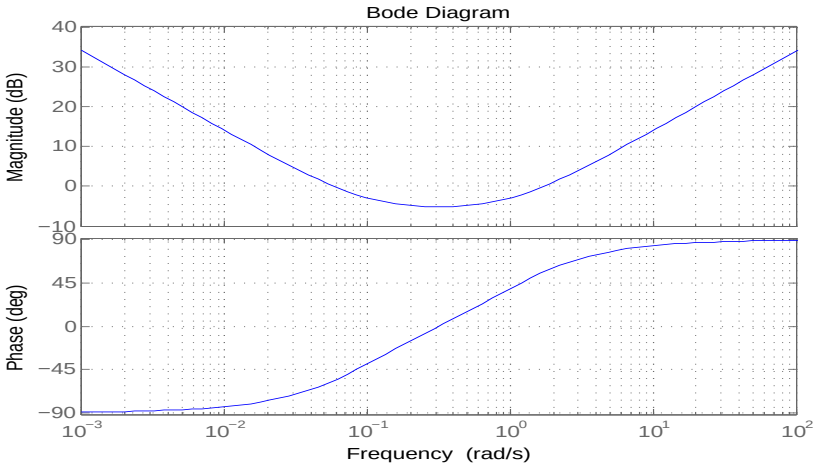


Figure 2.5: Bode plot for $K_3(s)$

3. Consider the feedback loop shown in Figure 3.1 below. Here

$$G(s) = \frac{1}{(s+2)^3}.$$

It is required to design a compensator $K(s)$ such that the following design specifications are satisfied by the closed-loop poles:

- The settling time is approximately 2 seconds.
- The response is oscillatory with a maximum overshoot of approximately 5%.

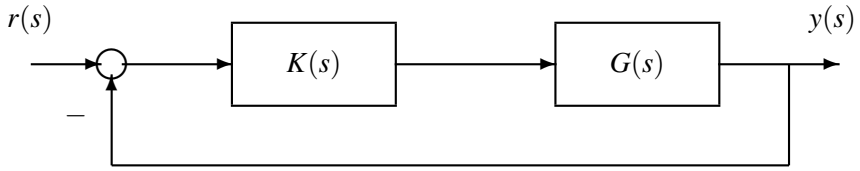


Figure 3.1

a) Find the location of the closed–loop poles that achieves the design specifications above. [5]

b) For this part, take $K(s) = k$.

i) Use the Routh–Hurwitz criterion to determine the range of values of k for which the closed–loop is stable. [5]

ii) Write down the closed–loop characteristic equation in terms of k. [5]

iii) Show that the design specifications cannot be achieved by any k. [5]

c) For this part, take $K(s)$ to be a PD compensator.

i) Design a PD compensator that achieves the design specifications. *Hint: You may want to choose the compensator zero to cancel one of the poles of* $G(s)$. [5]

ii) For this compensator, use the final value theorem to find the steady–state error when $r(t)$ is a unit step. [5]